Exact minimum measurement dependence for violating the CHSH inequality

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Statistical independence of the measurement settings is one of the two assumptions whose relaxation evades Bell's theorem; quantifying how much must be relaxed turns the superdeterminism objection from a qualitative suspicion into a number. We study local deterministic hidden-variable models for the CHSH scenario in which the hidden variable may depend on the settings, and we measure that dependence by $F = \max_{ab} \mathrm{TV}(p(\cdot \mid a,b) \,\|\, \rho)$, the worst-case total-variation distance between the conditional source $p(\lambda \mid a,b)$ and a fixed baseline source ρ. We prove $S \le 2 + 8F$ for *every* local CHSH model — on any hidden-variable space, with any source distribution and deterministic or stochastic responses — and show the bound is tight on $[0, 1/N]$ at each fixed $N \ge 4$ (on $[0, \frac{1}{4}]$ in the supremum over N). For N-state *uniform-source* models we compute the exact maximum $S_{\max}(N, F)$: it is piecewise linear in F and is given explicitly for $N = 2, \ldots, 8$ by exhaustive exact arithmetic. Within this class the minimum fine-tuning required to reach the Tsirelson value $2\sqrt{2}$ is $F^* = (\sqrt{2}-1)/\min(N,4)$; the floor $F^* \ge (\sqrt{2}-1)/4$ behind it holds for *all* local models. It saturates at $N = 4$, so adding hidden states beyond four does not lower the cost — the bottleneck is the four CHSH correlators, not the size of the hidden source. We further prove that the no-signalling constraint changes nothing for $N \ge 3$: $F^*_{\mathrm{NS}}(N) = F^*(N)$, with explicit algebraic witnesses at every N, and that no two-state model can violate Bell under no-signalling at any fine-tuning level. The minimal fine-tuning is therefore achievable without observable signalling. No-signalling is silent at the algebraic maximum as well: the unique NS box with $S = 4$ is the PR box, and the minimum fine-tuning for a local no-signalling decomposition of it is exactly the signalling cap $\lceil N/4 \rceil / N$ for all $N \ge 3$.

I. INTRODUCTION

Bell's theorem [1], in the Clauser–Horne–Shimony–Holt (CHSH) form [2], is the cleanest statement of quantum nonlocality: every local hidden-variable model satisfies $|S| \le 2$, while quantum theory reaches the Tsirelson bound $S = 2\sqrt{2}$ [3] and the algebraic maximum is 4. The theorem rests on two assumptions — locality (factorisability) and statistical independence of the settings from the hidden variable ("free choice"). Relaxing either one evades the theorem. The first assumption itself decomposes into parameter independence and outcome independence [4, 5], and local hidden-variable models are equivalently characterised by the existence of a joint probability for all four outcomes [6]. This paper quantifies the second escape route: *how much* may a local model's hidden variable depend on the measurement settings, and what is the minimum amount required to reproduce a given CHSH value?

The question converts the qualitative objection "superdeterministic models are conspiratorial" into a quantitative benchmark. If the required fine-tuning is small, the loophole is cheap and Bell's theorem says little about the real world; if it is non-trivial, every local explanation must bite hard. Earlier work established that only a modest relaxation suffices. Brans showed already in 1988 that Bell's theorem does not eliminate fully causal — i.e. measurement-dependent — hidden-variable models [7]; the resulting measurement-dependence ("superdeterminism") programme was developed by the later works below and reappraised recently in Ref. 8. Hall showed that a local deterministic model of singlet-state correlations can be built by giving up a fraction of measurement independence, with a cost of $M_{\mathrm{singlet}} = 2(\sqrt{2}-1)/3 \approx 0.276$ in his pairwise L^1 metric M [9]; Friedman *et al.* derived tight relaxed CHSH inequalities for arbitrary per-observer measurement dependence [10]; Hall and Branciard quantified the cost in mutual-information bits under different causal structures [11]. More recent work has developed "exact minimum measurement dependence" programs in Hall-type metrics, including trade-offs with hidden information [12] and, most recently, exact closed forms for CHSH and GHZ–Mermin inequalities [13, 14]; a total-variation-based bound for drifting preparation ensembles has also appeared [15]. Bacciagaluppi, Hermens and Leegwater argued that some measurement-independence violations amount to "signalling in principle" and that Bell's theorem holds if no-signalling is imposed [16].

We work in a complementary metric. For a model with hidden variable λ and conditional sources $p(\lambda \mid a,b)$, we define the fine-tuning cost as the worst-case total-variation distance of any conditional source from a fixed baseline source ρ,

$$F = \max_{ab} \mathrm{TV}\big(p(\cdot \mid a,b) \,\|\, \rho\big), \qquad \mathrm{TV}(p\|\rho) = \tfrac{1}{2}\sum_{\lambda} |p(\lambda \mid a,b) - \rho(\lambda)|, \tag{1}$$

$F = 0$ is exactly Bell's statistical-independence assumption; F measures, per settings block, how far the source

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must be bent toward the settings. The metric is deliberately not information-theoretic: it is a direct measure of the deviation of the joint distribution $p(\lambda, a, b)$ from the product form, in the same space where Bell's assumptions live.

Our main results are:

1. **A model-independent bound.** For every local CHSH model — on any hidden-variable space, with any source distribution and deterministic or stochastic responses — $S \le 2 + 8F$. The bound is tight on the interval $[0, 1/N]$ for every $N \ge 4$ (Sec. IV A).

2. **Exact curves.** For N-state models with a uniform source, $S_{\max}(N, F)$ is an exact piecewise-linear function of F, given in closed form for $N = 2, \ldots, 8$ by exhaustive exact arithmetic; the cap $F(S{=}4; N) = \lceil N/4 \rceil / N$ holds for all N (Secs. IV B and IV D).

3. **The minimum fine-tuning for the Tsirelson value.** For uniform-source models, $F^*(N) := \min\{F : S_{\max}(N, F) \ge 2\sqrt{2}\} = (\sqrt{2} - 1)/\min(N, 4)$: 0.20711 at $N = 2$, 0.13807 at $N = 3$, and $(\sqrt{2} - 1)/4 \approx 0.103553$ for all $N \ge 4$. The cost *saturates* at $N = 4$: more hidden states do not make the loophole cheaper; and the floor $F^* \ge (\sqrt{2} - 1)/4$ is model-independent, holding for every local model (Secs. IV A and IV C).

4. **No-signalling is silent.** Imposing full no-signalling on the observed marginals does not change the minimum for any $N \ge 3$: $F^*_{\rm NS}(N) = F^*(N)$, with explicit algebraic witnesses at every N; the same silence holds at the algebraic maximum, $F^*_{\rm NS}(S{=}4; N) = \lceil N/4 \rceil / N$, the signalling-allowed cap, since the unique NS box with $S = 4$ is the PR box (Sec. V D); and at $N = 2$, no fine-tuning level permits a Bell violation under no-signalling (Sec. V).

5. **An explicit witness.** A four-state model attaining $S = 2\sqrt{2}$ at exactly $F^* = (\sqrt{2} - 1)/4$, verified as an exact identity in $\mathbb{Q}(\sqrt{2})$ (Sec. VI).

We are careful about the strength of these claims. This is not a refutation of Bell's theorem, and we do not claim it as a breakthrough: the qualitative statement that "only a little fine-tuning suffices" is already in the literature [9–11]. What is new — no prior use was found in arXiv-indexed literature through 2026-08-23, a completed sweep we describe in Sec. VII C — is (i) the exact closed form $F^* = (\sqrt{2} - 1)/\min(N, 4)$ in the per-setting TV metric, with the $N = 4$ saturation, the exact curves and cap, and the model-independent bound $S \le 2 + 8F$; (ii) the no-signalling sub-case pinned to the same closed form for all $N \ge 3$, with explicit witnesses, plus the sharp two-state impossibility theorem and the algebraic-maximum cap $F^*_{\rm NS}(S{=}4; N) = \lceil N/4 \rceil / N$ (the PR box is the unique NS box with $S = 4$) — a precise quantitative answer, in the observable-marginal sense, to the "signalling in principle" question of Ref. 16; (iii) the explicit finite witness at exactly F^*; and (iv) the one-sided causal-structure value $F^*_{\rm os} = (\sqrt{2} - 1)/2$, flat in N, and the explicit metric-conversion bounds of Sec. VII E.

II. MODEL AND DEFINITIONS

A. The finite model

Fix the CHSH scenario: settings $a, b \in \{0, 1\}$, outcomes $x, y \in \{\pm 1\}$, correlators $E[ab] = \langle x\, y \rangle_{a,b}$, and

$$S = E[00] + E[01] + E[10] - E[11] \;=\; \sum_{ab} \sigma_{ab}\, E[ab], \quad (2)$$

with $\sigma_{ab} = +1$ for $ab = 00, 01, 10$ and $\sigma_{11} = -1$, the ab being ordered as $00, 01, 10, 11$. A *finite local model* with N hidden states is specified by:

- a uniform source $\rho(\lambda) = 1/N$ on $\lambda = 1, \ldots, N$;
- for each joint setting (a, b), a *conditional source* $p[ab, \lambda] \ge 0$ with row sum $\sum_\lambda p[ab, \lambda] = 1$. The settings may bias which source states occur — this is the measurement dependence;
- deterministic responses $u[a, \lambda], v[b, \lambda] \in \{\pm 1\}$ for Alice and Bob.

The correlators are $E[ab] = \sum_\lambda p[ab, \lambda]\, u[a, \lambda]\, v[b, \lambda]$, and the fine-tuning cost is Eq. (1). We write

$$S_{\max}(N, F) \;=\; \sup S \quad (3)$$

over all such models with fine-tuning at most F.

We also study the *no-signalling (NS) sub-case*: the observed marginals must not depend on the distant setting, i.e. for every a, $\sum_\lambda p[ab, \lambda]\, u[a, \lambda]$ is independent of b, and symmetrically for Bob. This asks whether the fine-tuning must leak into observable signalling, or whether a local model can hide it.

B. Product patterns and the parity lemma

For each λ define the *product pattern* $q_\lambda = (q_{00}, q_{01}, q_{10}, q_{11})$ with $q_{ab}(\lambda) = u[a, \lambda] v[b, \lambda]$. The product telescopes,

$$q_{00}\, q_{01}\, q_{10}\, q_{11} = (u[0, \lambda] u[1, \lambda])(v[0, \lambda] v[1, \lambda]) = +1, \quad (4)$$

so only the 8 *even-parity* patterns occur. Each has CHSH value $C(q) := q_{00} + q_{01} + q_{10} - q_{11} \in \{\pm 2\}$: four patterns have $C = +2$ — the *miss-type* patterns, each of which differs from σ at exactly one coordinate —

$$\begin{aligned} M_A &= (-1, +1, +1, -1), \quad M_B = (+1, -1, +1, -1), \\ M_C &= (+1, +1, -1, -1), \quad M_D = (+1, +1, +1, +1). \end{aligned} \quad (5)$$

and the other four are their negatives, with $C = -2$. The sign vector $\sigma = (+,+,+,-)$ itself has product -1, so *no even-parity pattern equals* σ: every state misses at least one of the four conditions $q_{ab} = \sigma_{ab}$ that would give $S = 4$. We call this the *parity lemma*; it is the key combinatorial fact behind the cap theorem (Sec. IV B).

III. EXACT REDUCTION TO A FINITE MAXIMUM

Two structural facts make $S_{\max}(N, F)$ exactly computable, and the exact statements below rest on them rather than on any numerical solver.

A. The per-row TV optimum

Lemma 1 (per-row TV optimum)**.** *Let $f \in \{\pm1\}^N$ have k entries equal to $+1$ and m entries equal to -1 ($k + m = N$), set $\mu = (k-m)/N$, and let $\mathcal{D}_F = \{p \in \Delta_N : \mathrm{TV}(p\|\rho) \le F\}$ with ρ uniform. Then*

$$\begin{aligned}\max_{p\in\mathcal{D}_F} p\cdot f &= \mu + 2\min(F, m/N),\\ \min_{p\in\mathcal{D}_F} p\cdot f &= \mu - 2\min(F, k/N),\end{aligned} \tag{6}$$

with both sides equal to μ in the degenerate cases $k = 0$ or $m = 0$.

Proof. Write $d = p - \rho$ (zero-sum). Then $p\cdot f - \mu = d\cdot f$, and two upper bounds apply. *(TV bound)* Since $|f_\lambda| = 1$, $|d\cdot f| \le \sum_\lambda |d_\lambda| = 2\,\mathrm{TV}(p\|\rho) \le 2F$. *(Mass bound)* Let $M^- = \sum_{f=-1} d_\lambda$; since $p \ge 0$, each of the m terms is $\ge -1/N$, so $M^- \ge -m/N$. Using $\sum_\lambda d_\lambda = 0$, $d\cdot f = (\sum_{f=+1} d_\lambda) - M^- = -2M^- \le 2m/N$; symmetrically $d\cdot f \ge -2k/N$. Combining gives the upper bound. Achievability is bang–bang: put all of $+\delta$ on one $+1$ entry ($p = 1/N + \delta$ there, legal since $\delta \le m/N \le (N-1)/N$) and spread $-\delta$ over $\lceil \delta N \rceil$ entries of the -1 set (each can absorb at most $1/N$, and $\lceil \delta N\rceil \le m$ for $\delta \le m/N$). Then $d\cdot f = 2\delta$ exactly with $\mathrm{TV}(p\|\rho) = \delta$. The minimum follows by moving mass from the $+1$ set to the -1 set, where the available mass is k/N. □

Lemma 1 says: a row can only be bent toward f at rate 2 per unit of TV, and it saturates once F reaches the fraction m/N of states that already point the "wrong" way. It is the only non-combinatorial ingredient in this paper; everything else is finite combinatorics on top of it.

B. The exact finite-maximum formula

Theorem 1 (exact finite maximum)**.** *For all $N \ge 2$ and all $F \in [0, 1]$,*

$$S_{\max}(N, F) \;=\; \max_{Q}\Big[\, S_0(Q) + \sum_{ab} g_{ab}(F)\Big], \tag{7}$$

where Q ranges over multisets of N even-parity patterns (from the eight of Sec. II), $S_0(Q) = \frac{1}{N}\sum_{\lambda\in Q} C(q_\lambda)$, $n_{ab}(Q) = \#\{\lambda \in Q : q_{ab}(\lambda) \ne \sigma_{ab}\}$, and

$$g_{ab}(F) = \begin{cases} 2\min(F,\, n_{ab}/N), & 0 < n_{ab} < N,\\ 0, & n_{ab} \in \{0, N\}.\end{cases} \tag{8}$$

There is no σ_{ab} sign in front of the min-terms: each row ab is optimised in its own σ_{ab}-direction *(toward $E[ab] = \sigma_{ab}$), so every contribution to S is positive.*

Proof sketch. Four steps. **(i) Vertices.** For fixed p, S is multi-affine in the response variables $u[a,\lambda], v[b,\lambda]$ (linear in each, since only products enter); its maximum over $\{\pm1\}^{4N}$ equals its maximum over $[-1,1]^{4N}$ and is attained at a vertex. Swapping the two maxima — legal because the p-feasible set (product of per-row TV balls with the simplex) is independent of (u, v), compact, and S is continuous in both factors, so the maximum over the product space is attained and the two orders agree — gives $S_{\max}(N, F) = \max_{(u,v)\in\{\pm1\}^{4N}} \mathrm{LP}(u, v)$, where LP maximises S over p with row sums 1 and per-row TV $\le F$.

(ii) Gauge reduction. S depends on (u, v) only through the products $q_{ab}(\lambda)$: $E[ab] = \sum_\lambda p[ab,\lambda]\, q_{ab}(\lambda)$. The per-state simultaneous sign flip $(u[\cdot,\lambda], v[\cdot,\lambda]) \mapsto (-u[\cdot,\lambda], -v[\cdot,\lambda])$ leaves all q's fixed, and any two vertices with the same pattern assignment have identical LP values. The maximum over 2^{4N} vertices reduces to a maximum over pattern assignments $Q = (q_1, \ldots, q_N)$, each even-parity.

(iii) Realisability. Every even-parity multiset is realisable by some vertex: given q_λ with product $+1$, set $u[0,\lambda] = +1$, $v[0,\lambda] = q_{00}$, $v[1,\lambda] = q_{01}$, $u[1,\lambda] = q_{00}q_{10}$; then $u[1,\lambda]v[0,\lambda] = q_{10}$ and $u[1,\lambda]v[1,\lambda] = q_{00}q_{01}q_{10} = q_{11}$ by even parity.

(iv) Row separability. For fixed Q, the LP objective $\sum_{ab}\sigma_{ab}\sum_\lambda p[ab,\lambda]q_{ab}(\lambda)$ and all constraints are separable in ab: each block $\{p[ab,\cdot]\}_\lambda$ is an independent probability vector subject to its own TV ball. Apply Lemma 1 to row ab with target direction $f_\lambda = \sigma_{ab}\, q_{ab}(\lambda)$; the "wrong-sign" count of Lemma 1 is then exactly $n_{ab}(Q)$ and $\mu_{ab} = \frac{1}{N}\sum_\lambda \sigma_{ab}q_{ab}(\lambda)$, giving $\max_{\text{row } ab}\ \ \sigma_{ab}\sum_\lambda p[ab,\lambda]q_{ab}(\lambda) = \mu_{ab} + g_{ab}(F)$. Summing over ab and using $\sum_{ab}\mu_{ab} = S_0(Q)$ gives Eq. (7). □

Consequences. For each N, $S_{\max}(N, F)$ is an *exact* piecewise-linear function of F: on each cell $F \in [k/N, (k+1)/N]$ every term $2\min(F, n/N)$ is affine, so $S_{\max}$ is the upper envelope of finitely many exact affine functions, computable in rational arithmetic over the $\binom{8+N-1}{N}$ multisets (36/120/330/792/1716/3432/6435 for $N = 2, \ldots, 8$). The slopes can only be $0, 2, 4, 6, 8$.

C. The round-robin construction

Let Q_N be the multiset formed by the first N entries of the periodic sequence $M_A, M_B, M_C, M_D, M_A, M_B, \ldots$, i.e. state i takes miss-type $M_{i \bmod 4}$. Then $S_0(Q_N) = 2$ (all patterns have $C = +2$) and each count $n_{ab}(Q_N)$ is $\lfloor N/4 \rfloor$ or $\lceil N/4 \rceil$ (the four counts differ by at most one). The construction attains

$$S = 2 + 8F \qquad \text{on } [0,\, 1/N] \text{ for every } N \geq 4, \tag{9}$$

since on that interval $F \leq 1/N \leq \lfloor N/4 \rfloor / N$ and all four rows are active. It is the single explicit family of response tables that gives the entire lower-bound curve for every $N = 2, \ldots, 8$ (Sec. IV D).

IV. RESULTS WITH SIGNALLING ALLOWED

A. The global bound and its generality

Theorem 2. *For every $N \geq 2$ and every model,*

$$S \;\leq\; 2 + 8F, \tag{10}$$

independently of N. The bound is tight on $[0, 1/N]$ for every $N \geq 4$.

Proof. For any model the identity $S = S_0 + \sum_{ab}(\sigma_{ab} E[ab] - \mu_{ab})$ holds, with μ_{ab} as in step (iv) of Theorem 1: summing its definition over ab gives $\sum_{ab} \mu_{ab} = \frac{1}{N} \sum_\lambda C(q_\lambda) = S_0$. Lemma 1 gives $\sigma_{ab} E[ab] - \mu_{ab} \leq 2 \min(F, n_{ab}/N) \leq 2F$ per row, so $S \leq S_0 + 8F$. Finally $S_0 = \frac{1}{N} \sum_\lambda C(q_\lambda) \leq 2$, since every even-parity pattern has $C(q) \in \{\pm 2\}$. Tightness on $[0, 1/N]$ for $N \geq 4$ is Eq. (9). □

The bound holds far beyond the finite uniform-source setting:

Proposition 1 (general-measure extension)**.** *Inequality* (10) *holds for any local CHSH model on an arbitrary probability space (Λ, ρ), with* any *source distribution ρ (uniformity is not used) and with deterministic or stochastic responses, where $F = \max_{ab} \mathrm{TV}(p(\cdot \mid a, b) \| \rho)$.*

Proof sketch. (i) *Deterministic responses.* Pointwise, q_λ has even parity, so $C(q_\lambda) \in \{\pm 2\}$ almost everywhere and $S_0 := \sum_{ab} \mu_{ab} = \int C(q)\, d\rho \leq 2$, where $\mu_{ab} := \int \sigma_{ab}\, q_{ab}\, d\rho$ is the measure-theoretic analogue of the finite-model definition. With ν_{ab} denoting the conditional source measure, $\nu_{ab}(\{\lambda\}) = p(\lambda \mid ab)$, the identity $\sigma_{ab} E[ab] - \mu_{ab} = \int \sigma_{ab} q_{ab}\, d(\nu_{ab} - \rho)$ holds, and for any f with $\|f\|_\infty \leq 1$, $\left|\int f\, d(\nu - \rho)\right| \leq \int |f|\, d|\nu - \rho| \leq 2\,\mathrm{TV}(\nu \| \rho)$; hence $|\sigma_{ab} E[ab] - \mu_{ab}| \leq 2F$ per row, and since $S = S_0 + \sum_{ab}(\sigma_{ab} E[ab] - \mu_{ab})$, we get $S \leq S_0 + 8F \leq 2 + 8F$.

(ii) *Stochastic responses.* Given λ, the four response distributions define a joint measure μ_λ on $\{\pm 1\}^4$ (local response independence given λ). The model equals the *deterministic* model on $\Lambda \times \{\pm 1\}^4$ with source $\rho'(\lambda, s) = \rho(\lambda) \mu_\lambda(s)$, conditional sources $p'_{ab}(\lambda, s) = p_{ab}(\lambda) \mu_\lambda(s)$, and responses s: the correlators match because $\sum_s \mu_\lambda(s)\, s_a t_b = \alpha_a(\lambda) \beta_b(\lambda)$ with $\alpha_a = E[x \mid a, \lambda]$, $\beta_b = E[y \mid b, \lambda]$. The TV cost is preserved *exactly*, since μ_λ factors out of the TV sum: $\frac{1}{2} \sum_{\lambda, s} |p'_{ab} - \rho'| = \frac{1}{2} \sum_\lambda |p_{ab}(\lambda) - \rho(\lambda)| \sum_s \mu_\lambda(s) = \mathrm{TV}(p_{ab} \| \rho)$. Case (i) applies to the enlarged model. □

Consequence. The lower bound $F^* \geq (\sqrt{2} - 1)/4$ derived below is a statement about *all* local models — every source distribution, deterministic or stochastic — not just finite uniform-source ones. No extension of the model class can lower the fine-tuning floor.

B. The cap: reaching $S = 4$

Theorem 3. *For all $N \geq 2$, the smallest fine-tuning that lets a local deterministic model reach $S = 4$ is*

$$F(S{=}4; N) \;=\; \frac{\lceil N/4 \rceil}{N}. \tag{11}$$

For $N = 2, \ldots, 8$ the values are $\frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{2}{5}, \frac{1}{3}, \frac{2}{7}, \frac{1}{4}$.

Proof. Since $|E[ab]| \leq 1$, $S = \sum_{ab} \sigma_{ab} E[ab] \leq \sum_{ab} |E[ab]| \leq 4$, with equality only if $E[ab] = \sigma_{ab}$ for all four ab. By Lemma 1, row ab can reach σ_{ab} only if $F \geq n_{ab}/N$ (for $0 < n_{ab} < N$; $n_{ab} = 0$ needs no budget), so $S = 4$ is achievable at budget F iff some Q has $n_{ab} \leq FN$ for all ab, i.e. $F(S{=}4; N) = \min_Q \max_{ab} n_{ab}/N$. By the parity lemma each λ misses at least one class, so $\sum_{ab} n_{ab} \geq N$ and hence $\max_{ab} n_{ab} \geq \lceil N/4 \rceil$ (necessity). The round-robin construction Q_N has $\max_{ab} n_{ab} = \lceil N/4 \rceil$, and at $F = \lceil N/4 \rceil / N$ every row can be supported entirely on its correct class (TV $= n_{ab}/N \leq F$), giving $S = 4$ (sufficiency). □

Note that the cap is *not* "1/4 for all $N \geq 4$"; e.g. $S_{\max}(5, \frac{1}{4}) = \frac{16}{5} + \frac{1}{2} = 3.7 < 4$. The frequently quoted statement "$S = 4$ requires $F \geq \frac{1}{4}$" remains true ($\lceil N/4 \rceil / N \geq \frac{1}{4}$ always) but is tight only when $4 \mid N$.

C. The minimum fine-tuning for the Tsirelson value

Theorem 4 (headline)**.** *For all $N \geq 2$,*

$$F^*(N) := \min\{F : S_{\max}(N, F) \geq 2\sqrt{2}\} \;=\; \frac{\sqrt{2} - 1}{\min(N, 4)}. \tag{12}$$

Thus $F^* = (\sqrt{2} - 1)/2 \approx 0.20711$ at $N = 2$, $(\sqrt{2} - 1)/3 \approx 0.13807$ at $N = 3$, and $(\sqrt{2} - 1)/4 \approx 0.103553$ for all $N \geq 4$.

Proof. For $N = 2, 3$ the exact curves (Theorem 5, proved below in Sec. IV D) give $S_{\max} = 2 + 4 \min(F, \frac{1}{2})$ and $2 + 6 \min(F, \frac{1}{3})$; solving $2 + 4F = 2\sqrt{2}$ and $2 + 6F = 2\sqrt{2}$ gives the stated values.

For $N \geq 4$: *lower bound* — by Theorem 2, $S \leq 2 + 8F$ for all F; since $2 + 8F$ is strictly increasing, $F < F^* \Rightarrow S < 2 + 8F^* = 2\sqrt{2}$ with $F^* = (\sqrt{2} - 1)/4$. *Achievability* — the round-robin Q_N suffices: its minimum count is $\min_{ab} n_{ab}(Q_N) = \lfloor N/4 \rfloor$, and one checks that $\lfloor N/4 \rfloor \geq (\sqrt{2} - 1)N/4$ for all $N \geq 4$: writing $N = 4k + r$, $r \in \{0, 1, 2, 3\}$, the inequality becomes $k(2 - \sqrt{2}) \geq (\sqrt{2} - 1)r/4$, i.e. $k \geq 0.177$, 0.354, 0.530 for $r = 1, 2, 3$ respectively — so $k \geq 1$ suffices in each case ($N \geq 5, 6, 7$). Hence at $F = (\sqrt{2} - 1)/4$ all four rows of Q_N are active and $S_{\max}(N, F^*) \geq S_0(Q_N) + 8F^* = 2 + 8(\sqrt{2} - 1)/4 = 2\sqrt{2}$. □

The same inequality $\lfloor N/4 \rfloor \geq (\sqrt{2}-1)N/4$ is the feasibility condition of the no-signalling construction in Sec. V.

D. Exact curves for $N = 2, \ldots, 8$

Theorem 5. *For $N = 2, \ldots, 8$, $S_{\max}(N, F)$ is exactly the following piecewise-linear function of F:*

$$
\begin{aligned}
N = 2 &: [0, \tfrac{1}{2}]\ 2 + 4F;\ \textit{then}\ 4\\
N = 3 &: [0, \tfrac{1}{3}]\ 2 + 6F;\ \textit{then}\ 4\\
N = 4 &: [0, \tfrac{1}{4}]\ 2 + 8F;\ \textit{then}\ 4\\
N = 5 &: [0, \tfrac{1}{5}]\ 2 + 8F;\ [\tfrac{1}{5}, \tfrac{2}{5}]\ \tfrac{16}{5} + 2F;\ \textit{then}\ 4\\
N = 6 &: [0, \tfrac{1}{6}]\ 2 + 8F;\ [\tfrac{1}{6}, \tfrac{1}{3}]\ \tfrac{8}{3} + 4F;\ \textit{then}\ 4\\
N = 7 &: [0, \tfrac{1}{7}]\ 2 + 8F;\ [\tfrac{1}{7}, \tfrac{2}{7}]\ \tfrac{16}{7} + 6F;\ \textit{then}\ 4\\
N = 8 &: [0, \tfrac{1}{4}]\ 2 + 8F;\ \textit{then}\ 4
\end{aligned}
\tag{13}
$$

The breakpoint values are rational, as the formula forces: $S_{\max}(5, \frac{1}{5}) = \frac{18}{5}$, $S_{\max}(6, \frac{1}{6}) = \frac{10}{3}$, $S_{\max}(7, \frac{1}{7}) = \frac{22}{7}$.

The upper bound is the exhaustive exact maximum of Eq. (7) over all $\binom{8+N-1}{N}$ multisets, in rational arithmetic; the lower bound is the round-robin construction Q_N (Sec. III C), which gives exactly these functions (Fig. 1). For general N, the first segment $2 + 8F$ on $[0, 1/N]$ ($N \geq 4$), the cap $\lceil N/4 \rceil / N$, and the headline value $S_{\max}(N, F^*) = 2\sqrt{2}$ are proved for all N; the full bend region for $N \geq 9$ is left to numerics (the enumeration cost grows as $\binom{N+7}{7}$; $N = 9$ already needs 11 440 multisets).

V. THE NO-SIGNALLING SUB-CASE

A. Definition and a gauge caveat

We impose full no-signalling on the observed marginals: for every a, $\sum_\lambda p[ab, \lambda]\, u[a, \lambda]$ is independent of b, and symmetrically for Bob. A subtlety matters here: NS constrains single-party marginals and is *not* invariant under the per-state sign-flip gauge of Theorem 1 — the products q_{ab} fix S but not the marginals. The witnesses below therefore specify full response tables (u, v), not just patterns; a naive "patterns-only" NS reduction fails.

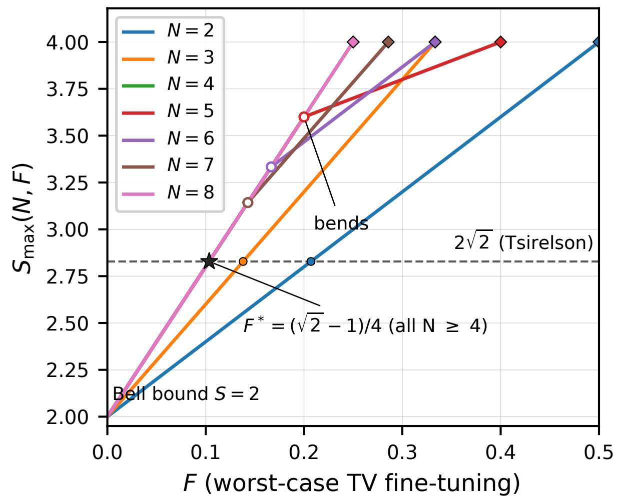

FIG. 1. Exact curves $S_{\max}(N, F)$ for $N = 2, \ldots, 8$ (Theorem 5), computed in exact rational arithmetic from Eq. (7). Every curve starts at the Bell value $S = 2$ with initial slope $2\min(N, 4)$. Open circles mark the bends at $F = 1/N$ for $N = 5, 6, 7$, where rows of the maximizing multiset exhaust their wrong-sign states and the slope drops; diamonds mark the caps $F(S{=}4; N) = \lceil N/4 \rceil / N$ (Eq. (11)), beyond which each curve is flat at $S = 4$. The star marks the saturation value $F^* = (\sqrt{2} - 1)/4$, at which all curves with $N \geq 4$ cross the Tsirelson value $2\sqrt{2}$ (dashed line); the dots are the $N = 2, 3$ crossings.

B. Two states: no violation at any fine-tuning level

Theorem 6. *Under full no-signalling, $S_{\mathrm{NS}}(2, F) = 2$ for* all *$F \in [0, \frac{1}{2}]$. No Bell violation is possible at any fine-tuning level with two hidden states.*

Proof (certificate). For $N = 2$, each row is parameterised by the free variable $t_{ab} = p[ab, \lambda_0] \in [\frac{1}{2} - F, \frac{1}{2} + F]$; S is affine in the four t's. The mechanism behind the flatness is that full NS pins row pairs on both sides at once: Alice's equations force $t_{a0} = t_{a1}$ whenever $u[a, \cdot]$ is non-constant, and Bob's force $t_{0b} = t_{1b}$ whenever $v[b, \cdot]$ is non-constant (e.g. $u[0, \cdot] \neq \text{constant} \Rightarrow t_{00} = t_{01}$). The four NS equations therefore collapse the four free variables to a small number of independent groups, and S is an affine function of those groups over a box of half-width F. With only one side constrained, half these pinning relations are absent and a slope-4 direction survives (the side finding below). Enumerating all $2^8 = 256$ response vertices and closing under the four NS equations by union-find reduces the free variables to groups; maximising the resulting affine S over the box gives, for each vertex, an explicit exact affine function $A + BF$ (the sign of B fixes which endpoint is maximal). In total, 1312 candidate affine functions are collected; every one satisfies $A + BF \leq 2$ for all $F \in [0, \frac{1}{2}]$ (checked at both endpoints), and $S = 2$ is attained for *every* $F \in [0, \frac{1}{2}]$ (e.g. the constant responses $u \equiv v \equiv +1$, for which

$E[ab] = 1$ and hence $S = 2$ at every feasible row). Hence $S_{\mathrm{NS}}(2, F) \equiv 2$. The certificate is exact (rational arithmetic) and re-runnable; see the Supplemental Material. [17] □

Side finding. Under *one-sided* NS only (Alice's marginals independent of Bob's setting), the bound is not flat — at $N = 2$, $S = 2 + 4F$ is reachable; the flatness genuinely uses both sides. (This is a constraint on the observed marginals; the different constraint that the hidden source itself be independent of Bob's setting — one-sided *measurement dependence* — is treated separately in Sec. VII B.)

C. All $N \geq 3$: no-signalling is silent

Theorem 7 (NS headline). *Under full no-signalling, for every $N \geq 3$,*

$$F^*_{\mathrm{NS}}(N) := \min\{F : S_{\mathrm{NS}}(N, F) \geq 2\sqrt{2}\} = \frac{\sqrt{2} - 1}{\min(N, 4)} = F^*(N). \tag{14}$$

Fine-tuning to the CHSH Tsirelson value is completely NS-silent: the minimum is identical with and without signalling, and it is attained by an explicit no-signalling model at every $N \geq 3$.

Proof. Lower bound. NS restricts the feasible set, so $S_{\mathrm{NS}}(N, F) \leq S_{\max}(N, F)$. For $N \geq 4$, Theorem 2 gives $S_{\max} \leq 2 + 8F < 2\sqrt{2}$ whenever $F < (\sqrt{2} - 1)/4$; for $N = 3$, the exact curve of Theorem 5, $S_{\max}(3, F) = 2 + 6 \min(F, \frac{1}{3})$, gives $S_{\max} < 2\sqrt{2}$ whenever $F < (\sqrt{2} - 1)/3$. Hence $S_{\mathrm{NS}}(N, F) < 2\sqrt{2}$ for all $N \geq 3$ with $F < (\sqrt{2} - 1)/\min(N, 4)$.
Upper bound — explicit algebraic witnesses.

$N \geq 4$**.** Let $d = F^* = (\sqrt{2} - 1)/4$. State $i = 0, \ldots, N - 1$ lies in class $j = i \bmod 4$, with class counts $n_j \in \{\lfloor N/4 \rfloor, \lceil N/4 \rceil\}$. The construction is:

- *Patterns:* state i has miss-type M_j (all four $C = +2$ patterns; Sec. II).
- *Response gauge* (per class j):

$$u[0, \cdot] = (+, +, -, -), \quad u[1, \cdot] = (-, +, +, -),$$
$$v[0, \cdot] = (-, +, -, -), \quad v[1, \cdot] = (+, -, -, -), \tag{15}$$

  i.e. the j-th entry of each tuple gives the response of states in class j. One checks that the resulting per-state products are exactly M_A, M_B, M_C, M_D for classes $0, 1, 2, 3$.

- *Rows:* $p[ab, i] = \frac{1}{N} + x[ab, i \bmod 4]$ with (all other $x = 0$)

$$\begin{aligned} &x[00, 0] = -d/n_0, && x[00, 1] = +d/n_1, \\ &x[01, 0] = +d/n_0, && x[01, 1] = -d/n_1, \\ &x[10, 1] = +d/n_1, && x[10, 2] = -d/n_2, \\ &x[11, 0] = +d/n_0, && x[11, 3] = -d/n_3. \end{aligned} \tag{16}$$

Verification is exact (rational and $\mathbb{Q}(\sqrt{2})$ arithmetic; certificate in the Supplemental Material):

- *Rows.* $n_r\, x[ab, r] = -d$ and $\sum_{j \neq r} n_j x[ab, j] = +d$ hold identically for each row, so every row has TV exactly d and is bang–bang in its σ_{ab}-direction. Feasibility $p \geq 0$ requires $d/n_r \leq 1/N \iff n_r \geq dN$ for all r — precisely the case-checked inequality $\lfloor N/4 \rfloor \geq (\sqrt{2} - 1)N/4$ of Sec. IV C.
- *NS.* Each NS equation — Alice's equating rows $a0$ and $a1$ (weighted by u_a), Bob's equating rows $0b$ and $1b$ (weighted by v_b) — reduces to $\sum_j n_j x[ab_L, j]\, w_j = \sum_j n_j x[ab_R, j]\, w_j$ for the response values w_j (the uniform parts cancel). Substituting Eqs. (15) and (21), each side is a sum of at most two terms; all four close explicitly. Alice a=0 (rows 00 vs 01, $u_0 = (+, +, -, -)$): $n_0(-d/n_0)(+1) + n_1(d/n_1)(+1) = -d + d$ versus $n_0(d/n_0)(+1) + n_1(-d/n_1)(+1) = d - d$. Alice a=1 (rows 10 vs 11, $u_1 = (-, +, +, -)$): $n_1(d/n_1)(+1) + n_2(-d/n_2)(+1) = d - d$ versus $n_0(d/n_0)(-1) + n_3(-d/n_3)(-1) = -d + d$. Bob b=0 (rows 00 vs 10, $v_0 = (-, +, -, -)$): $n_0(-d/n_0)(-1) + n_1(d/n_1)(+1) = 2d$ versus $n_1(d/n_1)(+1) + n_2(-d/n_2)(-1) = 2d$. Bob b=1 (rows 01 vs 11, $v_1 = (+, -, -, -)$): $n_0(d/n_0)(+1) + n_1(-d/n_1)(-1) = 2d$ versus $n_0(d/n_0)(+1) + n_3(-d/n_3)(-1) = 2d$. Each identity is $0 = 0$ or $2d = 2d$, independent of the class counts n_j; the same closure was verified exactly for all $N = 4, \ldots, 64$.
- *Value.* Each row contributes $\mu_{ab} + 2d$, so $S = S_0 + 8d = 2 + 8F^* = 2\sqrt{2}$, with $\mathrm{TV}(p_{ab} \| \rho) = d$ for all ab and NS holding. Hence $F^*_{\mathrm{NS}}(N) \leq F^*$ for all $N \geq 4$.

$N = 3$**.** In $\mathbb{Q}(\sqrt{2})$, set $d_3 = (\sqrt{2} - 1)/3$; the patterns are $\{M_A, M_B, M_D\}$ on states $1, 2, 3$, with response gauge

$$u[0, \cdot] = (-, -, +), \quad u[1, \cdot] = (+, -, +),$$
$$v[0, \cdot] = (+, -, +), \quad v[1, \cdot] = (-, +, +), \tag{17}$$

and rows (entries are $\frac{1}{3}$ plus d_3 times the coefficient shown)

$$\begin{array}{c|ccc} & \lambda_1 & \lambda_2 & \lambda_3 \\ \hline 00 & -1 & +\frac{1}{2} & +\frac{1}{2} \\ 01 & +\frac{1}{2} & -1 & +\frac{1}{2} \\ 10 & -\frac{1}{2} & +\frac{1}{2} & 0 \\ 11 & +\frac{1}{2} & +\frac{1}{2} & -1 \end{array} \tag{18}$$

Exact $\mathbb{Q}(\sqrt{2})$ arithmetic gives $S = 2 + 6d_3 = 2\sqrt{2}$, per-row TV $= (d_3, d_3, d_3/2, d_3)$, and all four NS residuals exactly zero. Hence $F^*_{\mathrm{NS}}(3) \leq d_3$. (The saved certified array — available from the authors, Supplemental Material Sec. C — differs from this printed gauge by the global flip $(u, v) \mapsto (-u, -v)$, which leaves every product

q_{ab} — hence S — and the setting-independence of each one-party marginal unchanged.)

Lower and upper bounds together give Eq. (14) for all $N \geq 3$, with explicit algebraic witnesses at every N — an existence-free statement. ☐

Remark (the whole NS curve on $[0, F^*]$.) Every constraint of the $N \geq 4$ construction is linear in d, so scaling d to any $F \in [0, F^*]$ gives a valid NS model with $\mathrm{TV}(p_{ab}\|\rho) = F$ and $S = 2 + 8F$ (feasibility holds throughout). With the global upper bound $S_{\mathrm{NS}} \leq S_{\max} \leq 2 + 8F$, this yields

$$S_{\mathrm{NS}}(N, F) = 2 + 8F \qquad \text{on } [0, F^*] \text{ for all } N \geq 4: \tag{19}$$

the NS curve coincides with the signalling-allowed curve up to the quantum point.

D. The algebraic maximum under no-signalling

The silence of Sec. V C extends to the algebraic maximum $S = 4$: the unique NS box that reaches it is the PR box, and the minimum fine-tuning of a local no-signalling decomposition of that box equals the signalling-allowed cap.

Lemma 2 (PR-box uniqueness). *A no-signalling CHSH box with $S = 4$ is uniquely the Popescu–Rohrlich (PR) box: at $(a, b) = (0, 0), (0, 1), (1, 0)$ it outputs $x = y$ with $P(+,+) = P(-,-) = 1/2$, and at $(1, 1)$ it outputs $x = -y$ with $P(+,-) = P(-,+) = 1/2$; all single-party marginals are uniform.*

Proof. $S = E[00] + E[01] + E[10] - E[11] = 4$ with $|E[ab]| \leq 1$ forces $E[ab] = \sigma_{ab}$ at all four corners. Since $x, y \in \{\pm 1\}$, $E[ab] = +1$ means $x = y$ almost surely, and $E[11] = -1$ means $x = -y$ almost surely. Write the NS marginals as $\alpha_a = P(x{=}+1 \mid a)$, $\beta_b = P(y{=}+1 \mid b)$ (NS makes them independent of the distant setting). The three $x = y$ corners give $\alpha_0 = \beta_0$, $\alpha_0 = \beta_1$, $\alpha_1 = \beta_0$; the $x = -y$ corner gives $\alpha_1 = 1 - \beta_1$. The first three yield a common value r; the last gives $r = 1 - r$, hence $r = 1/2$, and the box is fully determined. ☐

Consequence. A fine-tuned local deterministic model that is no-signalling and reaches $S = 4$ is *exactly* a local decomposition of the PR box: $P(x, y \mid a, b) = \sum_\lambda p[ab, \lambda]\, \mathbf{1}[x = u[a, \lambda]]\, \mathbf{1}[y = v[b, \lambda]]$ with per-row $\mathrm{TV}(p(\cdot \mid ab)\|\rho) \leq F$. The question is therefore: what is the minimum TV cost of a local decomposition of the PR box, as a function of the number N of hidden states? (The PR box is NS but nonlocal — it has no such decomposition at $F = 0$; the question is how much measurement dependence must be imported to make it local.)

Theorem 8. *For every $N \geq 3$,*

$$F^*_{\mathrm{NS}}(S{=}4; N) := \min\{F : S_{\mathrm{NS}}(N, F) \geq 4\} = \frac{\lceil N/4 \rceil}{N}. \tag{20}$$

No-signalling neither raises nor lowers the signalling-allowed cap of Theorem 3. The case $N = 2$ is excluded: by Theorem 6, $S_{\mathrm{NS}}(2, F) = 2$ for all F, so $S = 4$ is unattainable there at any fine-tuning level.

Proof. Lower bound. NS restricts the feasible set, so $S_{\mathrm{NS}}(N, F) \leq S_{\max}(N, F)$ and hence $F^*_{\mathrm{NS}}(S{=}4; N) \geq F(S{=}4; N) = \lceil N/4 \rceil / N$ by Theorem 3; no NS-specific argument is needed, since the parity-lemma necessity behind that cap ($S = 4 \Rightarrow E[ab] = \sigma_{ab}$ for all ab, so each row must be supported on its correct class, forcing $F \geq n_{ab}/N$ with $\max_{ab} n_{ab} \geq \lceil N/4 \rceil$) does not use NS.

Upper bound — closed-form construction. Use the round-robin patterns of Sec. III C: state i carries miss-type $M_{i \bmod 4}$, with class j of size $n_j \in \{\lfloor N/4 \rfloor, \lceil N/4 \rceil\}$. The response gauge is the base realisation per type with one whole class flipped — class D if $N \bmod 4 \in \{0, 1, 2\}$, class C if $N \bmod 4 = 3$ — which leaves every product q_{ab} (hence the patterns and S) unchanged and only fixes the NS marginals. Each row is then supported on its correct class C_{ab} — the class that misses ab carries mass 0 — so $E[ab] = \sigma_{ab}$ and $S = 4$. Writing $a, b, c, d = n_0, n_1, n_2, n_3$, the row masses on the four classes are, in the flip-D case,

$$\begin{array}{c|cccc} & A & B & C & D \\ \hline 00 & 0 & b/N & \frac{1}{2} - b/N & \frac{1}{2} \\ 01 & b/N & 0 & \frac{1}{2} - b/N & \frac{1}{2} \\ 10 & a/N & \frac{1}{2} & 0 & \frac{1}{2} - a/N \\ 11 & a/N & \frac{1}{2} & \frac{1}{2} - a/N & 0 \end{array} \tag{21}$$

(distributed uniformly within each class), and in the flip-C case the same display with b replaced by d and the roles of C and D swapped; all entries are non-negative for every $N \geq 3$. Two facts close the argument. **(NS automatic.)** Substituting the gauge into the four NS residuals, each reduces to the difference of two identical sums (e.g. Alice's $a{=}0$ residual is $(b/N + (\frac{1}{2} - b/N) - \frac{1}{2}) - (b/N + (\frac{1}{2} - b/N) - \frac{1}{2})$), so all four vanish identically: no NS equation constrains the free entries beyond the row sums. **(TV at or below the cap.)** Expanding the absolute values, each per-row TV is the maximum of two branch values; in the flip-D case,

$$\begin{aligned} \mathrm{TV}_{00} &= \max(a,\ N/2 - d)/N, \\ \mathrm{TV}_{01} &= \max(b,\ N/2 - d)/N, \\ \mathrm{TV}_{10} &= \max(N/2 - b,\ c)/N, \\ \mathrm{TV}_{11} &= \max(N/2 - b,\ d)/N. \end{aligned} \tag{22}$$

Since every class count is $\leq \lceil N/4 \rceil$, it remains to check the $N/2 -$ (class) branches: for $N = 4k + r$, $r \in \{0, 1, 2\}$, each is $\leq k + 1 = \lceil N/4 \rceil$ (equality at $r = 0, 2$), and in the flip-C case ($N = 4k + 3$) all four rows land at exactly $\mathrm{TV} = (k + 1)/N = \lceil N/4 \rceil / N$. Hence $\mathrm{TV}(p(\cdot \mid ab)\|\rho) \leq \lceil N/4 \rceil / N$ for all ab and all $N \geq 3$. ☐

Explicit witnesses (exact; certificate in the Supplemental Material). At $N = 3$ (class sizes $1, 1, 1, 0$, flip C), the per-state rows are $p[00, \cdot] = (0, \frac{1}{2}, \frac{1}{2})$, $p[01, \cdot] = (\frac{1}{2}, 0, \frac{1}{2})$,

$p[10,\cdot] = p[11,\cdot] = (\frac{1}{2},\frac{1}{2},0)$: $S = 4$, per-row TV 1/3, all four NS residuals zero, so $F^*_{\rm NS}(S{=}4;3) = \frac{1}{3}$. At $N = 4$ (class sizes $1,1,1,1$, flip D), the per-state rows are $p[00,\cdot] = (0,\frac{1}{4},\frac{1}{4},\frac{1}{2})$, $p[01,\cdot] = (\frac{1}{4},0,\frac{1}{4},\frac{1}{2})$, $p[10,\cdot] = (\frac{1}{4},\frac{1}{2},0,\frac{1}{4})$, $p[11,\cdot] = (\frac{1}{4},\frac{1}{2},\frac{1}{4},0)$: $S = 4$, per-row TV 1/4, NS residuals zero — an explicit local decomposition of the PR box at cost 1/4, so $F^*_{\rm NS}(S{=}4;4) = \frac{1}{4}$. At $N = 5$ (class sizes $2,1,1,1$, flip D), the class-mass rows are $p[00,\cdot] = (0,\frac{1}{5},\frac{3}{10},\frac{1}{2})$, $p[01,\cdot] = (\frac{1}{5},0,\frac{3}{10},\frac{1}{2})$, $p[10,\cdot] = (\frac{2}{5},\frac{1}{2},0,\frac{1}{10})$, $p[11,\cdot] = (\frac{2}{5},\frac{1}{2},\frac{1}{10},0)$: $S = 4$, per-row TV $\le \frac{2}{5}$, NS residuals zero, so $F^*_{\rm NS}(S{=}4;5) = \frac{2}{5}$.

Hiding the fine-tuning from the marginals never raises the price of reaching the algebraic maximum: no-signalling is silent at $S = 4$ as well as at $S = 2\sqrt{2}$, and the cost is the same cap $\lceil N/4\rceil/N$ as with signalling allowed.

VI. EXPLICIT WITNESS AT THE TSIRELSON POINT

The $N = 4$ optimal point of the signalling-allowed problem, verified by direct arithmetic and as an exact identity in $\mathbb{Q}(\sqrt{2})$ (Supplemental Material), is: with

$$F^* = \frac{\sqrt{2}-1}{4} \approx 0.1035534,$$
$$a = \tfrac{1}{4} - F^* \approx 0.1464466, \quad b = \tfrac{1}{4} + F^* \approx 0.3535534, \tag{23}$$

the conditional sources $p[ab,\lambda]$ (rows $=$ ab in order $00, 01, 10, 11$; columns $= \lambda = 1,\ldots,4$) are

$$\begin{array}{c|cccc} & \lambda_1 & \lambda_2 & \lambda_3 & \lambda_4 \\ \hline 00 & \frac{1}{4} & a & b & \frac{1}{4} \\ 01 & a & \frac{1}{4} & b & \frac{1}{4} \\ 10 & b & \frac{1}{4} & a & \frac{1}{4} \\ 11 & b & \frac{1}{4} & \frac{1}{4} & a \end{array} \tag{24}$$

with deterministic responses

$$\begin{aligned} u[0,\cdot] &= (+,-,-,-), \quad u[1,\cdot] = (+,+,+,-), \\ v[0,\cdot] &= (+,+,-,-), \quad v[1,\cdot] = (-,-,-,-). \end{aligned} \tag{25}$$

Direct recomputation (independent of any optimisation) gives $S = E[00] + E[01] + E[10] - E[11] = 2\sqrt{2}$ and $F = \max_{ab} \frac{1}{2}\sum_\lambda |p[ab,\lambda] - \frac{1}{4}| = (\sqrt{2}-1)/4$ exactly. A local, deterministic, finite model with a *uniform* source and only $\approx 10.4\%$ settings-conditioned fine-tuning (TV) thus reaches the maximal quantum CHSH violation exactly. (This is the signalling-allowed witness; the no-signalling witness of Theorem 7 has a different gauge and rows, since NS is not gauge-invariant.)

VII. DISCUSSION

A. Interpretation

The fine-tuning cost decreases with N but *saturates at* $N = 4$: adding hidden states beyond four does not lower the minimum fine-tuning needed to reach $2\sqrt{2}$. The bottleneck is the *four CHSH correlators*, not the size of the hidden source — a four-state source already spends its budget optimally across the four (a, b) blocks, and more states just dilute without reducing the per-correlator cost. (The qualitative observation that enlarging the hidden-variable space does not reduce the required relaxation was made earlier for mutual-information-metric singlet models [18]; our saturation is its exact closed-form counterpart in a different metric and target.) By Proposition 1 the floor $F^* \ge (\sqrt{2}-1)/4 \approx 0.1036$ is model-independent: *any* local CHSH model, on any hidden-variable space, with any source distribution and deterministic or stochastic responses, that reaches the maximal quantum violation must fine-tune at least one conditional source by at least $(\sqrt{2}-1)/4 \approx 10.4\%$ in TV in its worst settings block.

The no-signalling results sharpen the picture further. The minimum fine-tuning for $S = 2\sqrt{2}$ is achievable *without observable signalling* for every $N \ge 3$ (Theorem 7), and the whole curve up to the quantum point coincides with the signalling-allowed one (Eq. (19) for $N \ge 4$; at $N = 3$ the common curve is $2 + 6F$, by the same scaling argument). The cost of measurement dependence is therefore not paid in observable marginals: a local model can hide its settings-dependence from the single-party statistics and still violate CHSH maximally quantumly, at the same $\approx 10.4\%$ price. This directly addresses the claim of Bacciagaluppi *et al.* that some measurement-independence violations are "signalling in principle" [16]: in this TV metric, the *minimal* fine-tuning for a maximal quantum CHSH violation is NS-silent (for $N \ge 3$). Our statement concerns observable marginals within a fixed model class; it neither refutes nor settles the causal-structure notion of "signalling in principle" of Ref. 16, which is a different question. The two-state case is the converse: under full NS, no fine-tuning at all can produce a Bell violation (Theorem 6) — with two states, hiding the fine-tuning from the marginals makes it useless.

B. One-sided measurement dependence

A natural structural restriction, studied by Hall and Branciard in their mutual-information programme [11], is the *one-sided* causal structure: the hidden source depends on exactly one setting, $p[ab,\lambda] = q[a,\lambda]$ (independent of b), with cost $F_{\rm os} = \max_a {\rm TV}(q(a)\|\rho)$; by the $a \leftrightarrow b$ symmetry of CHSH, the Bob-only variant has identical values throughout. Rows then come in pairs, $p[00] = p[01] = q_0$, $p[10] = p[11] = q_1$, and

$$\begin{aligned} S &= \sum_\lambda q_0(\lambda)\, f^A(\lambda) + \sum_\lambda q_1(\lambda)\, f^B(\lambda), \\ f^A(\lambda) &= q_{00}(\lambda) + q_{01}(\lambda), \\ f^B(\lambda) &= q_{10}(\lambda) - q_{11}(\lambda). \end{aligned} \tag{26}$$

each f^A, f^B taking values in $\{-2, 0, +2\}$. The key structural fact: for every even-parity pattern, *exactly one* of $f^A(\lambda), f^B(\lambda)$ is non-zero — each hidden state can actively bend only *one* of the two source rows. That single fact halves the slope of the loophole and removes the saturation phenomenon.

Lemma 3 (one-sided row optimum)**.** *Let $f \in \{-2, 0, +2\}^N$ have k entries equal to $+2$, z entries equal to 0, and m entries equal to -2, and set $\mu = 2(k-m)/N$. Then*

$$\begin{aligned} h(f, F) &:= \max\{p \cdot f : \ p \in \Delta_N, \ \mathrm{TV}(p\|\rho) \le F\} \\ &\le \ \mu + 2F + 2\min(F, m/N) \ \le \ \mu + 4F, \end{aligned} \tag{27}$$

with the exact value a piecewise-linear function of F (water-filling over the three rate classes $+2, 0, -2$) given in the Supplemental Material.

Proposition 2. *For every $N \ge 2$,*

$$\begin{aligned} F^*_{\mathrm{os}}(N) &:= \min\{F : S_{\mathrm{os}}(N, F) \ge 2\sqrt{2}\} \\ &= \frac{\sqrt{2}-1}{2} \ \approx 0.20711, \end{aligned} \tag{28}$$

flat in N from the start, and the one-sided global bound $S_{\mathrm{os}}(N, F) \le 2 + 4F$ holds for all N and all F.

Proof sketch. Lower bound. Fix the responses and let k_A, m_A (resp. k_B, m_B) count the entries of f^A (resp. f^B) equal to ± 2. By the key structural fact, $k_A + m_A + k_B + m_B = N$. Lemma 3 bounds each row, and a four-case analysis of whether the two row bounds are above or below 2 — using $\mu_A + \mu_B = 2 - 4(m_A + m_B)/N$, which forces the intercept down whenever both rows are active — gives $S_{\mathrm{os}}(N, F) \le 2 + 4F$ for all N, F (exact case analysis in the Supplemental Material). Hence $S_{\mathrm{os}}(N, F) < 2\sqrt{2}$ for $F < (\sqrt{2}-1)/2$. *Upper bound.* Take $\lceil N/2 \rceil$ states of a pattern with $f^A = +2$ and $\lfloor N/2 \rfloor$ states of one with $f^B = +2$; then row A has $(k, m, z) = (\lceil N/2 \rceil, 0, \lfloor N/2 \rfloor)$ and row B the mirror, both in the slope-2 regime of Lemma 3, so $S_{\mathrm{os}}(N, F) \ge 2 + 4\min(F, \lfloor N/2 \rfloor/N)$. Since $(\sqrt{2}-1)/2 < \lfloor N/2 \rfloor/N$ for all $N \ge 2$ (an exact $\mathbb{Q}(\sqrt{2})$ identity, checked in the certificate), the construction reaches $S = 2 + 4F_{\mathrm{os}} = 2\sqrt{2}$ at $F_{\mathrm{os}} = (\sqrt{2}-1)/2$. □

Consequences. The one-sided cost equals the full-structure value at $N = 2$ exactly: adding hidden states never helps, so in cost terms the one-sided structure is worth throwing away every state beyond two. The penalty factor relative to the full causal structure is *exactly* 2 for $N \ge 4$ (and 3/2 at $N = 3$), since $F^*_{\mathrm{os}}(N)/F^*(N) = \min(N, 4)/2$. The cap to $S = 4$ under one-sided dependence is $\lceil N/2 \rceil/N$ (proved): exactly twice the full-model cap $\lceil N/4 \rceil/N$ for $N \equiv 0, 3 \pmod 4$, and a factor $\ge \frac{3}{2}$ (tending to 2) for $N \equiv 1, 2 \pmod 4$. Hall and Branciard's mutual-information metric gives the same qualitative ordering — one-sided ≈ 0.128 bits versus causal ≈ 0.080 bits, a factor ≈ 1.6 — whereas the TV metric gives the factor exactly 2; the two metrics are incommensurable, so no cross-metric ordering of the absolute numbers follows (Sec. VII E). The exact one-sided curves $S_{\mathrm{os,max}}(N, F)$ for $N = 2, \ldots, 8$ are certified in the Supplemental Material: for even N the curve is the single segment $\min(2 + 4F, 4)$ (proved for *all* even N), and for odd N it bends once at $F = \lfloor N/2 \rfloor/N$ (proved for $N = 3, 5, 7$; the bend line for odd $N \ge 9$ is conjectured).

C. Relation to prior work

Table I positions this work relative to the closest prior literature (all entries independently verified against arXiv; a completed 2015–2026 sweep with full search log is described in the Supplemental Material).

TABLE I. Comparison of fine-tuning metrics and values at the CHSH Tsirelson point $S = 2\sqrt{2}$. "Ret." = fraction of independence retained.

Work	Metric	Value at $S = 2\sqrt{2}$
Hall [9]	pairwise L^1 distance $M \in [0, 2]$ between setting-pair sources; NS, continuous λ	$M_{\mathrm{singlet}} = \frac{2}{3}(\sqrt{2}-1) \approx 0.276$ (ret. ≈ 0.862)
Hall & Branciard [11]	mutual information $I(X, Y : \Lambda)$ in bits, fixed causal structures	causal ≈ 0.080; retrocausal ≈ 0.046; one-sided ≈ 0.128; superdet = 2 bits
Friedman *et al.* [10]	per-observer Hall-type parameters, combined relaxations	tight general bounds (no single number)
Takakura *et al.* [12]	Hall's M + hidden information $H = 1 - \max_\lambda p(\lambda)$	$S \le 2 + \min\{3M, \frac{3}{4}M + 2H, 2\}$, necessary & sufficient
Pal *et al.* [15]	ensemble-divergence δ_{ens} (TV-type) for drifting preparations; MI preserved	$\lvert S\rvert \le 2 + 6\delta_{\mathrm{ens}}$ (not claimed tight)
Alai [13]	Hall-type variational M + detector efficiency η_{eff}	closed form $M(S, \eta)$; at $\eta = 1$: $(\sqrt{2}-1)/3 \approx 0.1381$
Alai [14]	Hall's M, $F := M/2$ (GHZ–Mermin)	$F(n) = R/[2(R+1)]$ ($R = 2^{\lfloor (n-1)/2 \rfloor}$) for $n = 3, \ldots, 13$; CHSH consistency value $(\sqrt{2}-1)/3$
This work	per-setting $\mathrm{TV}(p(\cdot\lvert ab)\|\rho)$ from a fixed baseline source	$F^* = (\sqrt{2}-1)/\min(N, 4) \approx 0.1036$ ($N \ge 4$), exact; NS-silent for $N \ge 3$

Three similar numbers — disambiguation. (i) Hall's L^1 convention: $M_{\text{singlet}} = \frac{2}{3}(\sqrt{2}-1) \approx \mathbf{0.276}$. (ii) Fraction-of-independence-lost convention ($M/2$): $(\sqrt{2}-1)/3 \approx \mathbf{0.1381}$ — Hall's $S = 2\sqrt{2}$ cost in that convention, and the $\eta = 1$ floor of Ref. 13. (iii) Our per-block TV metric: $F^* = (\sqrt{2}-1)/4 \approx \mathbf{0.1036}$. These come from different metrics on different model classes; no ordering of the numbers follows, and none contradicts any other. Hall's metric is pairwise (no fixed baseline) and his setting has no free conditional source; ours frees $p(\cdot \mid ab)$ against a fixed uniform baseline and allows (optionally forbids) signalling. Our TV is a different, non-information-theoretic measure of the same statistical-independence violation; explicit conversion bounds between the conventions for the *same* model, and the identification of our model class with the general (retrocausal) causal structure of Ref. 11, are worked out in Sec. VII E.

Novelty, stated as a search result. Within arXiv-indexed literature through 2026-08-23, no prior use was found of (i) the per-setting TV-from-a-fixed-source metric, (ii) the exact closed forms $F^* = (\sqrt{2}-1)/\min(N,4)$ with $N = 4$ saturation together with the exact curves and cap $\lceil N/4 \rceil / N$, or (iii) an NS-silence result of this kind. This is stated as a search result, not "proven novel": coverage was arXiv-centric ($\sim$16 papers read at full or abstract depth, the rest at catalogue level); paywalled journal content not mirrored on arXiv, non-arXiv repositories, and books/theses are not covered. The closest prior work is Ref. 13 — exact minimum measurement dependence for CHSH via LP with $\mathbb{Q}(\sqrt{2})$ certificates, but a Hall-type metric plus detector-efficiency degrees of freedom; its $\eta = 1$ floor at $2\sqrt{2}$, $(\sqrt{2}-1)/3 \approx 0.1381$, is a different number from our $F^* = (\sqrt{2}-1)/4 \approx 0.1036$ because the metrics differ — no contradiction. Together with Refs. 12, 14, and 15, this shows that "exact minimum measurement dependence" is an active 2026 front; our contribution is a specific metric with clean closed forms, plus the NS sub-case, which none of them covers.

D. Implications for proposed superdeterministic models

The model-independence of Proposition 1 means that the floor $F^* \geq (\sqrt{2}-1)/4 \approx 10.4\%$ is a price attached to the *statistics* of any local account of the Tsirelson value, independent of the causal story told for the settings-dependence. Several explicit models have been proposed in the superdeterminism literature, and all fall on the priced side of this bound. Brans's fully causal model [7] — in which the hidden variable determines both outcomes and settings and reproduces the singlet correlations for arbitrary spin-1/2 directions, as described in Ref. 9 — is a direct instance of our model class with a settings-dependent conditional source $p(\lambda \mid a, b)$; if it reaches $S = 2\sqrt{2}$, some settings block must carry $\geq 10.4\%$ TV. 't Hooft's cellular-automaton program [19] claims that any quantum model can be mimicked to arbitrary finite accuracy by classical deterministic dynamics, with Bell's theorem inapplicable because the realistic state distribution depends on the measurement basis; reaching $S = 2\sqrt{2} - \varepsilon$ in such a construction forces $F \geq (\sqrt{2}-1)/4 - O(\varepsilon)$. In Palmer's discretised Hilbert-space models [20, 21] the hidden variable determines the actual apparatus settings, so conditioning on observed settings restricts λ to a proper subset of its domain; since total variation is non-increasing under coarse-graining, any level of description that locally reproduces $2\sqrt{2}$ forces $F \geq 10.4\%$ at the fine-grained level as well — "invisible to coarse-graining" arguments cannot lower the cost below the floor. The most explicit recent construction we examined, Garza and Hance's conservation-law model [22], keeps a fixed uniform baseline source while measurement error removes a settings-dependent band from the hidden-variable domain; at its best-fit parameter it reaches $|S| \approx 2.826$, within 0.08% of the Tsirelson value, with worst-block TV of $\approx$ 17–29% — a factor $\approx$ 1.7–2.8 above the floor — and the inequality $S \leq 2 + 8F$ holds across its parameter space (verified numerically; Supplemental Material). [23] We also note a point of self-consistency in that construction: their correlator (their Eq. (14)) satisfies $|E(\theta; \Delta L)| \leq 1$ for all θ if and only if $\Delta L \geq \frac{1}{2}$, and their model already takes $\Delta L \geq 0.5$ — nothing they claim lies in the $|E| > 1$ regime. We stress what the bound does and does not say: it is compatible with every one of these models, and it rules out none of them; it fixes a quantitative price floor for the loophole, so that "a little fine-tuning" in the sense of $< 10.4\%$ TV per settings block is insufficient, by theorem, for any local reproduction of the maximal quantum violation. Whether such a deviation is physically plausible is a separate question, on which the experimental and model-selection literature to date weighs against superdeterministic causal structures [24, 25].

E. Metric comparison

The fine-tuning cost can be measured in four conventions (Table II). They all measure the same statistical-independence violation and differ only in (i) whether a fixed baseline ρ is used and (ii) the L^1-vs-half-L^1-vs-entropy functional — which is exactly why explicit conversion bounds exist between them.

Conversion bounds. For every local CHSH model and every baseline ρ:

1. $M_{\text{model}} \leq 2F$, by the triangle inequality, $\mathrm{TV}(p_{ab} \| p_{a'b'}) \leq \mathrm{TV}(p_{ab} \| \rho) + \mathrm{TV}(p_{a'b'} \| \rho) \leq 2F$;
2. $F \leq M_{\text{model}} + \mathrm{TV}(p_{\text{avg}} \| \rho)$, by $\mathrm{TV}(p_{ab} \| \rho) \leq \mathrm{TV}(p_{ab} \| p_{\text{avg}}) + \mathrm{TV}(p_{\text{avg}} \| \rho)$ and taking the maximum over ab;
3. no bound $F \leq f(M_{\text{model}})$ holds for a *fixed* external ρ: if all four rows equal some $q \neq \rho$, then $M_{\text{model}} = 0$ while $F = \mathrm{TV}(q \| \rho) > 0$.

TABLE II. The fine-tuning conventions of this work and the closest literature. $p_{ab} = p(\cdot \mid a, b)$, $\mathrm{TV}(p\|q) = \frac{1}{2}\|p-q\|_1$, $p_{\mathrm{avg}} = \frac{1}{4}\sum_{ab} p_{ab}$.

Metric	Definition
F (this work)	$\max_{ab} \mathrm{TV}(p_{ab}\|\rho)$: worst case from a *fixed* baseline ρ
M_{model}	$\max_{ab,a'b'} \mathrm{TV}(p_{ab}\|p_{a'b'})$: pairwise, no baseline
M (Hall [9])	$\max_{ab,a'b'} \|p_{ab} - p_{a'b'}\|_1 = 2M_{\mathrm{model}}$
$I(X, Y; \Lambda)$ (H&B [11])	$H(\Lambda) - \frac{1}{4}\sum_{ab} H(\Lambda \mid a, b)$, in bits

Bound (i) is tight: for both $N = 4$ witnesses (Secs. V and VI) the six pairwise TVs take values in $\{\delta, 2\delta\}$ with $\delta = F^*$, so $M_{\mathrm{model}} = 2F^*$ exactly — but it is not saturated by every F^*-optimal witness: the $N = 3$ NS witness of Theorem 7 has $M_{\mathrm{model}} = \frac{3}{2}F^*$ (slack $F^*/2$).

Our witnesses in the other conventions. The F^*-optimal $N = 4$ witnesses cost $M_{\mathrm{model}} = 2F^* = (\sqrt{2}-1)/2 \approx 0.207$ in the pairwise-TV convention, i.e. Hall's full-L^1 value $M = \sqrt{2}-1 \approx 0.414$ — which in his "fraction of independence retained" reads $1 - M/2 = (3-\sqrt{2})/2 \approx 0.793$ — and they carry $I(X, Y; \Lambda) \approx 0.056$ bits; the $N = 3$ NS witness carries ≈ 0.055 bits.

Different metrics select different optima. Our TV-optimal witness is not M-optimal: Hall's bound $S \le \min\{2 + 3M, 4\}$ — whose upper-bound proof needs only local determinism, not no-signalling (re-proved in the Supplemental Material) — at $M = \sqrt{2}-1$ allows up to $S = 3\sqrt{2}-1 \approx 3.24$, whereas our witness sits at $2\sqrt{2} \approx 2.83$; nor is it MI-optimal, its ≈ 0.056 bits exceeding the relevant floor of ≈ 0.046 bits. Different metrics select different optimal models; this is why the three numbers of the disambiguation above (0.276/0.138/0.104) coexist without contradiction.

Model-class identification. Ref. 11 minimizes $I(X, Y; \Lambda)$ under fixed causal graphs. Its causal structure imposes the additional factorization constraint $p(x, y \mid \lambda) = p(x \mid \lambda)\, p(y \mid \lambda)$, which our closed-form witnesses do *not* satisfy (verified exactly in $\mathbb{Q}(\sqrt{2})$ for every witness of this paper); equivalently, since Ref. 11 proves its causal floor ≈ 0.080 bits is the minimum over that class, no causal model can reach $2\sqrt{2}$ at ≈ 0.056 bits. Our model class — free $p(\lambda \mid a, b)$ against a fixed ρ — is therefore H&B's *general* (retrocausal) structure, for which the relevant floor at $S = 2\sqrt{2}$ is $I_{\mathrm{R}} \approx 0.046$ bits, not the causal floor $I_{\mathrm{C}} \approx 0.080$; our witnesses' ≈ 0.056 bits sit above it, as they must.

A sharper bound in our variable. Hall's $S \le 2 + 3M$ becomes $S \le 2 + 12F$ when written in our variable ($M = 2M_{\mathrm{model}} \le 4F$), versus our direct $S \le 2 + 8F$: TV from a fixed ρ yields the sharper CHSH bound because all four rows share one center.

Proposition 3 (aggregation robustness). *(i) For every local CHSH model — any source, deterministic or stochastic responses — $S \le 2 + 8F_{\mathrm{avg}}$, where $F_{\mathrm{avg}} = \frac{1}{4}\sum_{ab} \mathrm{TV}(p(\cdot \mid ab)\|\rho)$. (ii) The $N = 4$ witness of Sec. VI has all four per-row TVs equal to $d = (\sqrt{2}-1)/4$, so $F^*_{\mathrm{avg}}(4) := \min\{F_{\mathrm{avg}} : S = 2\sqrt{2}$ achievable$\} = d = F^*(4)$.*

Proof sketch. (i) The per-row estimate $|\sigma_{ab}E[ab] - \mu_{ab}| \le 2\mathrm{TV}(p(\cdot \mid ab)\|\rho)$ behind Proposition 1 uses only the *row's own* TV; summing the four rows with row-specific TVs gives $S \le S_0 + 2\sum_{ab} \mathrm{TV}(p(\cdot \mid ab)\|\rho) = S_0 + 8F_{\mathrm{avg}} \le 2 + 8F_{\mathrm{avg}}$. (ii) An exact $\mathbb{Q}(\sqrt{2})$ certificate (Supplemental Material) re-derives the witness arithmetic: the four per-row TVs are all exactly d, so its average-TV cost is d, and (i) gives the reverse inequality. □

The headline is therefore insensitive to the aggregation choice at $N = 4$: fine-tuning cannot be made cheaper by spreading it across the setting blocks rather than concentrating it in one. What is *not* claimed: $F^*_{\mathrm{avg}}(N)$ for $N > 4$ has not been computed; only the model-independent floor $(\sqrt{2}-1)/4$ and the $N = 4$ equality are proved.

F. Scope, caveats, and verification

The exact *curves* are certified for $N = 2, \ldots, 8$ (exhaustive enumeration); for $N \ge 9$ the first segment, the cap, and the headline F^* are proved for all N, while the full bend region is left to numerics. The exact finite-N statements are certified for the *uniform* source; Proposition 1 shows the lower bound $F^* \ge (\sqrt{2}-1)/4$ does not depend on the source distribution or on determinism, so no extension can lower the floor. Finally, F^* is a mathematical minimum within the model class — it is not compared to any physical prior on how much fine-tuning a real source could have.

Every result in this paper is checkable without trusting the prose: the exact statements carry re-runnable exact-arithmetic certificates (rational and $\mathbb{Q}(\sqrt{2})$, no floating point; exit code 0 iff every assertion passes), and an independent LP pipeline reproduces the same values to $\le 1.8 \times 10^{-15}$. The full verification record, including how to re-run each check, is in the Supplemental Material.

VIII. CONCLUSION

We quantified the price of the measurement-dependence escape route for Bell/CHSH in a per-setting total-variation metric. The price has a closed form: $F^* = (\sqrt{2}-1)/\min(N, 4)$, saturating at four hidden states; the global bound $S \le 2+8F$ is model-independent and tight; the cap to the algebraic maximum is $\lceil N/4 \rceil / N$; the no-signalling constraint is exactly silent about the minimum for all $N \ge 3$ — at both the Tsirelson value and the algebraic maximum, where the unique NS box is the PR box — with explicit algebraic witnesses, while at $N = 2$ it forbids any violation; and the one-sided causal structure pays exactly twice the full-structure cost, flat in N. The superdeterminism loophole, in this metric, costs a concrete $\approx 10.4\%$ of TV per settings block — and that cost can be hidden from the observed marginals. The systematic comparison with Hall's M and the conventions of Refs. 11 and 13 is given in Sec. VII E, and the generalisation of the program beyond CHSH — CGLMP, Mermin, and the Peres–Mermin square — is carried out in the companion paper [26], which identifies the structural conditions (parity frustration, the number K of extremal conditions, the minimal miss set) under which the saturation form survives. Natural next steps are extending the exact-curve certificates to $N \geq 9$ and to the open I3322 inequality.

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[To be added.]

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